EXECUTION-PATH QUALIFICATION AND REALIZED COSTS IN SPECULATIVE DECODING ON CONSUMER SYSTEMS

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ABSTRACT

Local speculative-decoding opportunity establishes neither output preservation nor integrated speedup. Fresh direct-ID gates qualify RTX serialization but leave B570 blocked. On the tested B570 stack, a target-only micro-batch-one predictor frozen before reserved live outcomes matches 72/72 complete generated-ID sequences within a 128-ID/EOG endpoint: 24 repeat-stable prompts, three runs each. Four prompts (12 runs) diverge from the reference, which matches 60/72. A separate same-state B570 intervention changes scores at 72/72 sampled forks but no immediate decisions; 18 later trajectories differ across six prompts. Separately frozen RTX proposal/verification programs predict 72/72 live runs on another 24-prompt set, while reference qualification remains 60/72. They share the numerical runtime and do not transfer B570's target-only hypothesis. A new directly qualified RTX serialized path is slower than target-only (0.646× target/serialized latency) under its instrumented post-prefill boundary. On 358H, an isolated NPU draft advantage does not survive the combined replay-plus-proposal interface. On 60 held-out M4 prompts, Guard and fixed speculation preserve 180/180 matched sequences; Guard reaches 1.3114× speedup, but fixed speculation is faster (1.3469×). The studies separate output qualification, bounded trajectory prediction and intervention, and same-path realized cost without locating a unique numerical cause.

1 INTRODUCTION

Cheap drafting and batched target validation motivate speculative decoding (Leviathan et al., 2023). Consumer hardware already shows gains and slowdowns without verification parallelism (Chordiya, 2026). We ask whether the intended execution path preserves output, whether frozen path predictions survive reserved outcomes, and whether local opportunity survives complete realized cost.

The original prospective RTX/B570 studies share HumanEval30, Qwen3-8B/0.6B Q8_0, greedy $k = 2$, three repetitions and a 90/90 direct-ID gate. RTX changes from 75/90 batched to 90/90 serialized; B570 changes from 60/90 fast to 78/90 serialized and remains blocked. These gates do not certify historical timing. A later independent RTX study freshly qualifies its own timer frontend and measures matched cost: serialization is slower. Qualification is necessary for a speed claim, not sufficient for a gain.

On the tested B570 stack, a target-only UB1 predictor selected at 36/36 discovery matches predicts 72/72 reserved runs: 24 repeat-stable prompts × three, four reference-divergent (12 runs), versus UB128 reference/live 60/72. A separate live intervention then tests identical fork states: scores change, sampled immediate decisions do not, and some later trajectories differ. On RTX, target-only H1 fails discovery; a separately frozen H2 program instead constructs and seals draft proposals and target trajectories before reserved live. Its 72/72 prediction equality coexists with 60/72 reference equality. These tests discriminate prediction, sampled intervention and qualification.

The four systems in Table 1 are independently frozen cases, not a matched hardware ranking. Heterogeneous placement, numerical path dependence and adaptive speculation are established (Ruiz y Mesa et al., 2026; Wang et al., 2026; Yuan et al., 2025; Gond et al., 2026; Liu et al., 2026a). Our three contributions are (1) fresh direct-ID execution-path qualification and matched cost on a new qualified RTX path, with historical comparisons kept separate; (2) frozen B570 target-only prediction, its bounded same-state intervention, and independent RTX proposal/replay prediction on a shared numerical runtime; and (3) realized 358H state-interface cost and a qualified M4 runtime test that finds no incremental Guard benefit over fixed speculation.

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2 Experimental Question and Measurement Discipline

Acceptance, isolated draft throughput and target-forward batching measure different boundaries. None alone measures the complete speculative loop. We use an output contract and an accounting lens to separate them.

2.1 A path-qualified output contract

A greedy speculative round verifies draft proposals, commits the matching prefix and a correction or bonus when budget permits, then removes inconsistent suffix state. Our strict contract requires committed generated IDs to equal the matched target-only greedy reference under the frozen configuration. This is an empirical property of model, runtime, path and workload. A failed pair blocks a workload-global optimization claim; a small passing probe only establishes eligibility for broader testing. Capped ID equality does not measure functional HumanEval correctness or pass@k.

Historical text-path agreement (H) compares arrays reconstructed from parsed CLI text. Direct-ID qualification (Q) compares captured IDs against the study's frozen reference. Frozen prediction equality (P) compares a sealed prediction with reserved live IDs. A predictor can match a reference-divergent path, so P does not imply Q. Same-state intervention separately compares scores, immediate decisions and subsequent trajectories; these endpoints must not be interchanged.

2.2 Evidence classes

Primary results answer a frozen question after its applicable gate; **secondary** results retain their registered or post-study follow-up scope. **Mechanism-supported** results use path controls or measured regions. **Diagnostic** values describe blocked or restricted paths. Historical RTX controls remain text-path secondary evidence and B570 timing diagnostic, irrespective of later direct-ID results.

2.3 Complete round costs

Let T_b denote matched target-only cost per output token, T_d^{real} the complete realized draft-subsystem time including interface replay, $T_{\text{target,round}}^{\text{real}}$ complete target work, and T_x residual synchronization, transfer and bookkeeping outside those regions. Count all committed output, including correction/bonus, as C. Then

$$S = \frac{T_b C}{T_d^{\text{real}} + T_{\text{target,round}}^{\text{real}} + T_x} \tag{1}$$

$$= \frac{C}{D_{\text{real}} + R_{\text{target}} + X}. \tag{2}$$

Terms in the second denominator are normalized by T_b; break-even requires $C > D_{\text{real}} + R_{\text{target}} + X$. These are accounting identities, not new profitability theory (Kim et al., 2026). Aggregate means and totals require consistent weights; total-time ratios need not equal prompt geometric means. Accepted drafts alone do not count correction/bonus output. Incompatible experiments cannot isolate an unmeasured component by subtraction.

2.4 A separate target-forward proxy

For context n and path e,

$$V_k^{\text{micro}}(n, e) = \frac{T_{\text{target forward}}(k, n, e)}{T_{\text{target decode}}(1, n)}. \tag{3}$$

This narrower microbenchmark omits drafting and may omit correction/bonus and state work; V_k^{micro} is not complete R_{target}. Isolated draft throughput likewise cannot substitute for stateful D_{real}. These boundaries guide diagnosis without supplying a universal cause.

3 Experimental Methodology and Platforms

3.1 Frozen systems and workloads

Campaign outcomes were frozen independently; their synthesis is retrospective. Table 1 describes historical hardware cases. Follow-ups have campaign-specific frontends, references and environments (appendix configuration table), rather than one setting per device. RTX's original direct-ID OS build is 26200; the new H1, H2 and latency studies use 26300. Driver versions alone do not identify driver bytes. Final comparisons process one request at a time. The target/draft pair is Qwen3-8B/0.6B Q8_0, except the 358H adapter's symmetric INT4 OpenVINO draft.

The representative M4 study selects 24 unseen Spec-Bench prompts across six groups (Xia et al., 2024). A separate preregistration selects 30 HumanEval tasks before input inspection (Chen et al., 2021); fixed $k = 2$ is primary and threshold variants secondary. Three repetitions and a 128-ID cap apply to the code studies. The prospective runtime studies retain their separate task splits and estimator in Section 7.

For direct-ID studies, "complete" means all committed/generated IDs through the first inclusive EOG or 128-ID cap, excluding prompt IDs and uncommitted proposals. Original B570 H1 reserved arrays all reach the cap; RTX H2 includes genuine earlier EOGs. Each follow-up uses its own reference and selected tasks. Exposure-informed discovery is distinct from reserved inference: audit and deterministic selection precede reserved execution, and predictions/tapes are sealed before reserved live. No-known-exposure means no record in the audited scope, not proof about unavailable host files. The appendix preserves selection, freezes, denominators and endpoint distributions.

Table 1. Historical case-study settings; follow-up campaign configurations are listed separately in the appendix. These are not a matched hardware comparison. All final target models are Qwen3-8B Q8_0. Greedy execution uses campaign-matched target-only references; historical M4/RTX/B570 use re-tokenized text, while 358H uses direct IDs.

System	Runtime / target backend	Draft model / placement	Evidence scope
MacBook Air M4	llama.cpp b10809 / Metal	Qwen3-0.6B Q8_0 / Metal	Representative 24Q and separate HumanEval30
Core Ultra 9 275HX + RTX 5070 Ti Laptop	llama.cpp b10839 / CUDA	Qwen3-0.6B Q8_0 / CPU or same GPU	Blocked fast path; text-path-agreeing controls
Core i5-13400 + Arc B570	llama.cpp b10839 / Vulkan0	Qwen3-0.6B Q8_0 / B570	Formal gate failure; later diagnostics
Core Ultra X7 358H + Arc B390	llama.cpp b10839 / Vulkan0; OpenVINO draft	Qwen3-0.6B symmetric INT4 / CPU or NPU	Closed Phase 3; direct stateful decomposition

Timing excludes loading and initial prefill. Native historical counters and the 358H complete-loop timer keep their original boundaries. The new RTX timer starts after synchronized target/draft prefill and before per-request sampler and generation-state work, ending at slot release; its instrumented CLI boundary is detailed in Section 5.4. Different runtimes, quantization and references preclude hardware-only causal comparisons.

3.2 Statistical estimators and uncertainty

Historical M4/RTX/B570 reduce three paired ratios by within-prompt GM; 358H uses their median. Headline speedups are prompt GMs with 95% prompt-bootstrap intervals. M4 HumanEval uses 100,000 resamples, the other hardware campaigns 10,000. New RTX latency instead takes each condition's three-time median, then target/serialized ratio and prompt GM. Its 10,000-draw prompt bootstrap preserves frozen seed, order and linear percentiles. M4 runtime has its own ordered streams. Repetitions and within-prompt decision events are correlated; they are not independent prompts. Intervals condition on the campaign and do not include full session, thermal or recalibration uncertainty.

3.3 Provenance and energy boundaries

An internal claim ledger maps canonical files and hashes to each study. Git object/file hashes identify retained bytes; server acknowledgements and host records supply the reported ordering, not a trusted external per-process clock. Anonymous derivatives preserve array values and rehash exported content, but cannot certify original remote chronology or omitted weights/runtime/compiler/driver bytes. Offline checks execute no models.

M4 energy measures package aggregate excluding separate DRAM; B570 measures adapter board energy. RTX mixed domains are not a common system boundary. No end-to-end 358H Phase-3 or runtime energy result is claimed. Historical Windows per-run provenance is weaker; 358H draft-source reconstruction remains `STRONGLY_SUPPORTED`, not exact historical identity. No platform energy ranking follows.

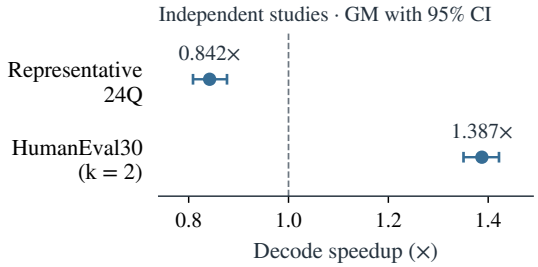


Figure 1. **Opposite profitability regimes on the same tested M4 stack**. Representative 24Q yields 0.842× [0.809, 0.877], while the separately preregistered HumanEval30 fixed-$k = 2$ study yields 1.387× [1.351, 1.422]. Points are campaign prompt geometric means of paired decode speedups; intervals are campaign-specific 95% prompt-bootstrap intervals. The dashed line marks parity. Both pass the recorded re-tokenized-text comparator (72/72 and 90/90 pairs); these are not direct generated-ID comparisons or functional HumanEval pass@k. These are separate studies, not a paired between-workload comparison; differing workloads and session conditions do not isolate workload-only causality.

4 Gains Do Not Translate Uniformly

Table 2 reports historical timing under recorded text-path agreement for M4/RTX/B570. Fresh direct-ID qualification appears separately in Section 5.1 and Figure 3; it does not convert those timings into strict-ID performance results. The separately divided new RTX row reports fresh qualification and matched instrumented latency (Section 5.4); it is a different campaign. A blocked path remains diagnostic.

Table 2. Headline decode outcomes. Speedup is baseline/speculative time under each frozen prompt estimator; brackets are 95% prompt-bootstrap CIs. Above the divider, historical M4/RTX/B570 rows show recorded text-path agreement, not direct generated-ID qualification. Below it, 358H rows show direct-ID sequence equality. The historical B570 60/90 is distinct from the fresh fast-path direct-ID 60/90 in Section 5.1. The original RTX/B570 qualification-only campaigns in Figure 3 have no matched latency. The new RTX block below uses its own fresh qualification and timer frontend; it does not certify historical timing. B570 timing remains diagnostic.

Path / workload	Speedup [95% CI]	Endpoint	Pairs	Faster prompts	Class
M4 representative 24Q	0.842 [0.809, 0.877]	Text-path	72/72	1/24	Primary
M4 HumanEval fixed-$k = 2$	1.387 [1.351, 1.422]	Text-path	90/90	30/30	Primary
CPU to RTX 5070 Ti Laptop	0.404 [0.390, 0.418]	Text-path	90/90	0/30	Secondary
RTX 5070 Ti Laptop to itself	0.650 [0.629, 0.673]	Text-path	90/90	0/30	Secondary
B570 serialized fallback	0.3087 [0.2927, 0.3246]	Text-path	60/90	0/30	Diagnostic
358H CPU draft	0.734 [0.693, 0.771]	Direct-ID	90/90	0/30	Primary
358H NPU draft	0.352 [0.333, 0.369]	Direct-ID	90/90	0/30	Primary
New RTX serialized latency	0.646 [0.630, 0.660]	Direct-ID	90/90	0/30	Matched cost

4.1 M4: outcomes in two separate studies

The representative frozen M4 sample produces 0.842× speedup (95% CI [0.809, 0.877]), with 72/72 text-path-equal pairs and only 1/24 prompts faster. The separately preregistered HumanEval fixed-$k = 2$ study produces 1.387× [1.351, 1.422], with 90/90 text-path-equal pairs and 30/30 prompts faster. Figure 1 retains both; the positive code-conditioned result does not replace or pool with the negative representative result.

The same M4 supports both profitable and unprofitable observed regimes that pass the historical text-path comparator. Profitability is therefore not a simple hardware capability property in these studies. These outcomes do not establish general M4 profitability or isolate unified memory as a cause of the profitable result. These are workload-conditioned observed regimes, not a controlled estimate of workload causality: workloads and session conditions differ. The earlier discovery effect size is not promoted as a confirmatory estimate. Historical MLX failures also do not negate the text-path-agreeing historical Q8_0 Metal result.

For primary fixed-$k = 2$ HumanEval, the package-efficiency ratio (baseline/speculative J per output token) is 1.013× [0.990, 1.034]. The interval includes parity; a package-energy improvement is not established. The registered $p_{\min} = 0.9$ condition has secondary energy evidence, discussed separately in the appendix. The preregistered corrected synthetic ceiling was not executed. We retain that deviation rather than treating the earlier synthetic diagnostic as its replacement.

4.2 Across stacks: negative and blocked outcomes

On RTX 5070 Ti Laptop, the serialized CPU-to-RTX control reaches 0.404× [0.390, 0.418], while the same-GPU draft control reaches 0.650× [0.629, 0.673]. Both pass 90/90 text-path comparisons and accelerate zero of 30 prompts. The higher speed of same-GPU placement shows that placement and associated integration costs matter, but its continued loss rules out transfer as the sole explanation.

On Arc B570, neither the fast path nor the serialized fallback supports a workload-global admissible optimization result. The latter's diagnostic speedup is 0.3087× [0.2927, 0.3246], with zero of 30 prompts faster. That interval describes timing on a partially mismatching path; its precision does not turn it into an admissible speed claim. The formal failure and stop rule remain the platform result. Board-energy observations likewise remain diagnostic and boundary-limited.

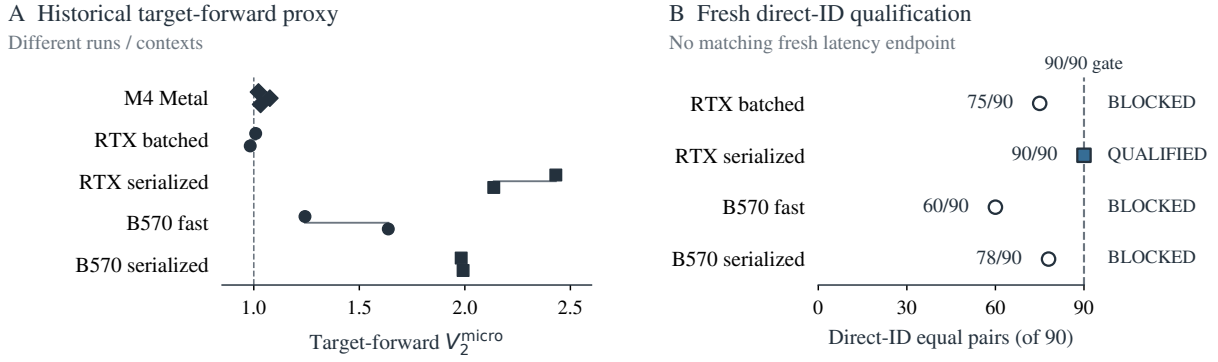
On Core Ultra X7 358H, the closed Phase-3 heterogeneous runner has direct-ID sequence equality on both CPU and NPU drafting: 90/90 pairs per path, with zero of 30 prompts faster. CPU achieves 0.734× [0.693, 0.771]; NPU achieves 0.352× [0.333, 0.369]. These direct-ID-qualified negative outcomes coexist with favorable isolated draft measurements. Section 5 investigates which measured boundaries account for the failed local promise.

5 From Local Opportunity to Realized Gain

The evidence proceeds from output qualification to frozen path prediction and bounded intervention, then same-path realized cost. Figure 2 retains the original B570 discovery/freeze/reserved sequence; subsequent studies use their own tasks and gates.

Stage	Design or comparator	Complete generated-ID endpoint
Discovery	12 prompts × 3 runs; select target-only UB1 as H1.	UB1 predicts the serialized trajectory in **36/36** runs.
Freeze	24 reserved prompts × 3 runs; fix 72 UB1 arrays before live outcomes.	No live reserved outcome is used to select or alter H1.
Live test	Compare each of 72 live serialized runs with both frozen paths.	UB1 predictor/live **72/72**; UB128 reference/live **60/72 (BLOCKED)**. Four prompts (12 runs) are reference-divergent.
Divergent case	HumanEval/134; first difference at generated-ID index 88.	Reference ID 914 → predictor/live ID 1946 in all 3 runs.

Figure 2. **B570 prospective target-partition prediction.** The 72 comparisons are repeat-stable executions of 24 distinct prompts, not 72 independent tasks. The same-set reference contrast and divergent example distinguish prediction of the observed blocked trajectory from restored qualification. H1's freeze and live ordering are reported in the appendix; complete arrays appear in the separate anonymous sequence supplement. This one-stack observational predictor is not a history-controlled live-loop intervention.



A: path shapes only; all 11 measured cells. B: open = BLOCKED; filled square = QUALIFIED.

Figure 3. **Historical target-forward proxy and fresh direct-ID qualification are separate evidence.** A: historical V_2^{micro} cells from different runs and contexts; shapes encode M4 reference (diamond), fast/batched (circle), and serialized (square) paths, with no qualification encoding. Segments span measured contexts, not uncertainty. The proxy is not complete R_{target} or end-to-end speedup. B: fresh HumanEval30 direct-ID sequence equality, with no matching fresh latency endpoint; only RTX serialization reaches the frozen 90/90 gate. Open circles mean BLOCKED; the filled square means QUALIFIED. The panels are not a matched cost–qualification experiment. Fresh direct-ID qualification does not retrospectively authorize historical timing. Results are path/runtime scoped; contexts and references differ.

5.1 Execution-path qualification

Figure 3A retains eleven historical target-forward proxy cells. Metal $V_2^{\mathrm{micro}} \approx 1.02$–$1.08$, near-step RTX batching, and B570 fast 1.24–1.64 versus serialized 1.983–1.992describe different runs and contexts. They omit integration, cannot replace complete R_{target}, and do not explain M4's separate-session contrast.

The original direct-ID studies each complete 270 executions and 180 comparisons over 30 HumanEval prompts × three, requiring 90/90 reference equality. RTX matches 75/90 batched and 90/90 serialized; only serialization qualifies. B570 matches 60/90 fast and 78/90 serialized; both remain blocked. Historical B570 text-path 60/90 and fresh fast direct-ID 60/90 are different comparisons. Repeat stability does not imply reference equality. These qualification-only campaigns have no matched latency; Figure 3A/B cannot form a speed/qualification tradeoff. The separate new RTX timing study below does not certify historical timing.

5.2 B570 prediction and sampled live intervention

A target-only harness first reproduces 90/90 UB128 reference sequences. H1 target-only UB1 matches 36/36 sequences on 12 exposed discovery tasks; its frozen tree selects H1, leaving H2 unrun. After conservative exposure exclusion, 24 reserved tasks are selected. H1 freezes before reserved inference and all 72 full predictions freeze before reserved live. Prediction/live agrees 72/72 over three stable runs per prompt, including four reference-divergent prompts (12 runs); the same-set UB128 reference agrees only 60/72 and remains *BLOCKED*. All these reserved arrays reach 128 IDs. The predictor could have failed on any reserved sequence; its success establishes predictive path sufficiency on this B570 stack, not a unique numerical cause. The earlier

serialized 78/90 gate has a different task set and reference comparison.

A separately frozen intervention uses 24 new reserved prompts × three. Each sampled checkpoint is the first verification block beginning with at least 64 committed IDs. Before UB1 versus UB128 target verification, the branches share byte-checked KV, prefix, position, fixed proposals and cloned sampler state. Raw scores change at 72/72 forks, but raw top-1 and actual immediate decisions change at 0/72; the 195 paired decision events are correlated within prompts. Subsequent full trajectories differ in 18/72 runs across 6/24 prompts.

The alternate branch restores the fork's draft state and continues with its UB128 target context; later proposals follow that branch's own live state. Thus the secondary continuation differences are not sampled-block immediate decision effects or proof that a one-block switch restores reference. These data show score/continuation effects under the frozen policy and no detected immediate effect at sampled states. They neither establish absence of partition effects throughout generation nor locate one operation or explain RTX. The appendix documents clone/R0 controls and offline limits.

5.3 RTX independent proposal and replay prediction

The separately frozen RTX H1 discovery tests target-only UB2 against native batched live and matches only 24/36, failing its 36/36 gate. No hypothesis is selected, its H2 interface is unavailable, and reserved execution is zero. This failure remains distinct from subsequent V2 work.

V2 implements independent draft-only P and target-only R processes. P constructs proposals using H2's own prior feedback; R performs verification, acceptance/correction/bonus and rollback. A fresh R_ONLY loads only target weights and consumes the sealed tape sequentially, computing its own decisions without future live answers. Original CLI live generates its own draft and imports no tape. Code, model-loading and RPC records support application/process/API independence; all programs share llama.cpp/GGML/CUDA numerical routines.

Discovery H2/live and H2/R_ONLY both pass 36/36. Following F0/F1/F2 freezes, complete predictions and tapes precede reserved live on another 24-task set × three: 288 executions across reference, H2, R_ONLY and live. H2/live, R_ONLY/live and H2/R_ONLY each match 72/72, whereas this study's UB1 reference/live matches 60/72. Four tasks (/17, /33, /14, /22) diverge in all repetitions. Live endpoints contain 63 earlier-EOG runs and nine capped runs; "complete" does not mean all reach 128. The recorded proposals and target path suffice to reconstruct these trajectories; H2 construction still uses drafting. This is independent program-level prospective reproduction, not direct transfer of B570 H1, an independent numerical implementation, or qualified batched performance.

5.4 Matched cost on a newly qualified RTX path

A separate arm builds and freezes an instrumented frontend, then obtains fresh serialized/reference direct-ID equality of 90/90 on 30 prompts × three × two conditions (180 timed runs). The prompt estimator is median target time divided by median serialized time, followed by a GM and fixed prompt bootstrap: 0.646× [0.630, 0.660], with 0/30 prompt ratios above one (Table 2). Serialization is slower within this qualified measured boundary.

The timer includes draft and target computation, per-request sampler/generation state, acceptance/correction/bonus, synchronization, state management and related logging/response work through slot release; loading, initial prefill and teardown are excluded. Committed-ID/debug logging is enabled. The empty-hook qualification does not bound all logging overhead below 1%. This new result neither replaces historical RTX 0.650 nor supplies a full batched-to-serialized speed tradeoff: no qualified batched timing exists in this same study.

5.5 Stateful realization

On 358H, isolated NPU draft throughput exceeds CPU at tested contexts, but both integrated direct-ID-qualified paths are slower than target-only. Complete committed progress is $C = 2.4631$ and 2.4584, including correction/bonus. Complete target work consumes 87.34% of CPU-path time. The NPU synchronous OpenVINO region combines full-prefix replay/reconstruction and proposal, consuming 58.35%; these costs are inseparable. Residual orchestration is 0.04% and 0.02%.

Figure 4 characterizes this interface, not NPU capability or a measured persistence benefit. Arithmetic totals give CPU $(D_{\text{real}}, R_{\text{target}}, X) = (0.4304, 2.9767, 0.00124)$ and NPU $(4.1251, 2.9435, 0.00130)$, yielding 0.7227× and 0.3477× in Equation 2; these are total-time ratios rather than Table 2's prompt GMs. The later persistent-state/wider-target intervention stops at qualification without formal runtime performance or energy.

6 From Measured Constraints to Runtime Decisions

Output preservation and realized cost constrain usable speculation independently. Original RTX serialization crosses its direct-ID gate; B570 improves agreement without crossing it. New RTX serialization freshly qualifies yet remains slower. The frozen B570 target-only predictor and independently orchestrated RTX replay reproduce reference-

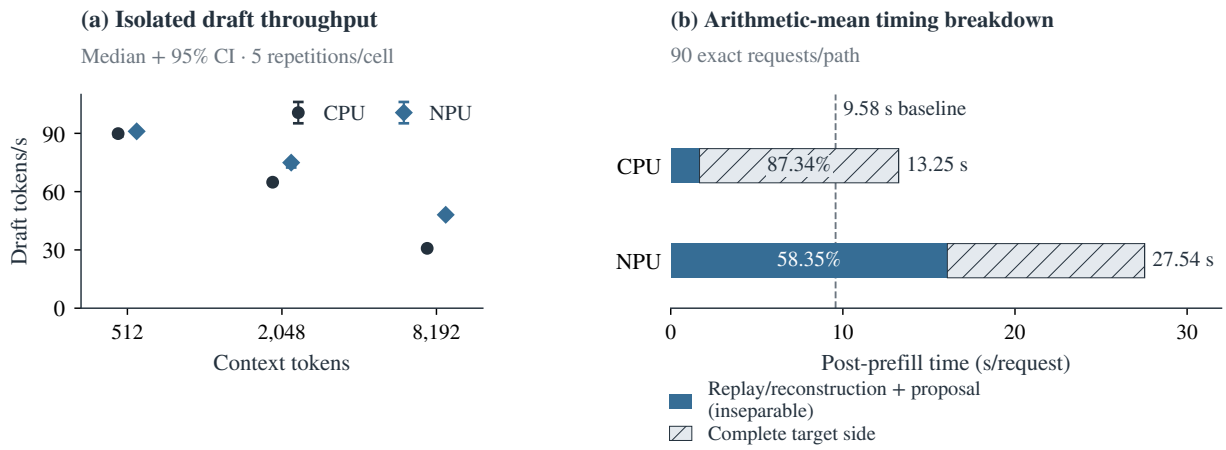


Figure 4. **Isolated draft advantage does not survive the tested stateful integration**. The NPU's 58.35% region is a single *full-prefix reconstruction/replay plus proposal* call; its components are inseparable and replay-only savings are unmeasured. (A) Phase-1 median steady-state draft throughput and 95% repetition-bootstrap intervals, five repetitions per cell. (B) Phase-3 arithmetic mean post-prefill seconds per request over 90 exact observations per path. CPU and NPU totals are 13.25 s and 27.54 s against a 9.58 s mean baseline. The CPU path is target-side dominated (87.34%). The 0.734× and 0.352× headlines use a separate prompt-GM estimator, not ratios of these arithmetic-mean bars. Initial prompt prefill is excluded. This diagnoses the tested interface, not intrinsic NPU capability or a measured state-persistence benefit. CPU replay plus proposal is 12.63%; NPU complete target work is 41.63%. Residuals are 4.81 ms (0.04%) and 5.06 ms (0.02%). Prompt-GM 95% CIs are [0.693, 0.771] and [0.333, 0.369]. The dashed line is the shared arithmetic-mean baseline; device marks are horizontally offset.

divergent paths, but prediction success does not qualify those paths. The B570 intervention further distinguishes changed scores and later trajectories from unchanged immediate decisions at sampled forks.

First qualify the intended path against its own reference. If blocked, test a precisely frozen path explanation with reserved trajectories and distinguish prediction from controlled state effects. Then measure complete realized cost on a qualified path. On 358H, an isolated draft-device advantage fails at the measured interface; lowering draft cost alone need not remove dominant target work. The separately frozen M4 runtime test below examines the practical limit of continuation control. These cases support this bounded evaluation sequence, not a common numerical cause or a cross-stack intervention chain.

7 Qualification and Realized-Cost Runtime Control

We examine qualification and continuation in separately frozen runtime studies. The 358H candidate stops at direct-ID qualification. On M4, a prospectively direct-ID-qualified path permits a test of a simple cost rule: Guard remains faster than target-only, but fixed speculation is faster in the post-study paired secondary comparison; no incremental control benefit is shown. These are complementary tests of the decisions in Section 6, not two successful optimization interventions.

7.1 An intervention stopped by qualification

The 358H component measurements motivated persistent draft state and a wider target execution setting. The implementation ran the target and draft backends in separate processes. Synchronous draft calls, including communication and state work, remained inside the target worker's post-prefill timer. The executable implementation still had to pass correctness qualification. CPU persistent-state restore/reconcile correctness could not be established under the tested representation; the Optimum NPU state-capable path was unavailable with the frozen dynamic-shape IR. These are implementation boundaries, not evidence that the devices cannot retain state.

The target setting `n_ubatch=2` (UB2) passed the small gate but failed full qualification. Correctness-only follow-up controls on Qwen3-8B Q8_0, llama.cpp b10839, and Vulkan0 reproduced different greedy tokens under UB1 and UB2 even with identical exact prefix tokens. Request-order and separate-process controls reproduced the target-only divergence. Replaying frozen proposals also reproduced the speculative divergence without live NPU draft infer-

ence. The latter depended on earlier target execution history; the controls found no position drift or specific rollback-boundary bug. The responsible floating-point or backend operation remains unidentified.

Under this frozen tested target stack, UB2 did not preserve the strict direct-ID output contract. The formal registry retained only the target reference, so the runtime study had no qualified intervention pair to evaluate. No formal R2 performance run occurred, and the follow-up diagnostics are invalid for performance claims. Qualification excluded an intervention motivated by the measured bottlenecks before its speed was evaluated.

7.2 A prospective cost guard on M4

The M4 experiment asks whether a frozen rule based on already-realized complete cost can control continuation while preserving exact output and profitable execution against target-only. It does not assume that adding control improves on fixed speculation. We retain the Qwen3-8B/0.6B Q8_0 pair on llama.cpp Metal, fixed $k = 2$, greedy generation, and a 128-token cap. Guard begins on the same speculative path as Always-Spec and can fall back permanently to target-only within a request. It changes neither proposal width nor device placement.

At each eligible completed round, Guard divides all elapsed charged generation cost by committed output progress C. This count includes the initial target seed and committed correction or bonus tokens, but excludes rejected proposals. Guard continues if this cumulative cost per token is at most $(1 - m)R_{\mathrm{ref}}$, a target-only reference frozen from calibration; otherwise it falls back. If a decision is reached with zero committed progress, Guard also falls back. Before each control decision, invalid candidate suffix state is removed from the target and draft contexts, preserving committed history for target-only continuation. The enclosing post-prefill clock includes generation-loop initialization, seed-token sampling, drafting, target work, synchronization, state reconciliation, control, and transition costs. Model loading, sampler/speculator setup, initial prompt prefill, and the speculative begin hook precede the timer; the result is not whole-request latency.

The reference is the median across calibration prompts of each prompt's median target latency divided by its median output-token count. The preregistered candidate grid varies the first decision round and margin. Selection minimizes calibration geometric-mean latency relative to the offline minimum of target-only and Always-Spec among exact candidates, with frozen tie-breaks. The selected rule uses margin 0 and first decides after 8 completed speculative rounds; it then decides after each later round while speculation continues. A request ending earlier makes no decision. The online policy receives neither task/family labels nor future output length, current-request target-only latency, or oracle outcomes.

Table 3. M4 per-prompt median post-prefill latencies. Each speculative condition has 180/180 direct-ID sequence equality. The first two rows report speedups. The offline oracle uses the lower target-only or Always-Spec median and is unavailable online; its row reports latency regret. The separated Guard/Always-Spec contrast is reviewer-requested post-study paired secondary analysis; values above one mean Guard is slower.

Latency ratio	GM	95% prompt CI
Target / Guard	1.3114	[1.2487, 1.3787]
Target / Always-Spec	1.3469	[1.2867, 1.4088]
Guard / offline oracle	1.0278	[1.0137, 1.0446]
Post-study paired secondary latency ratio		
Guard / Always-Spec	1.0271	[1.0130, 1.0436]

7.3 Held-out evaluation

Historical M4 outcomes informed the study design. Prospective here means that the policy was frozen before the reserved prompt outcomes were measured, not that the design was uninformed by earlier campaigns. IDs marked as previously executed or outcome-exposed in the frozen inventory were excluded before deterministic hash-ranked selection. Disjoint splits contain 40 calibration prompts, 16 qualification prompts, and 60 held-out prompts, with equal HumanEval and Spec-Bench counts within each split. The policy and reference were frozen before qualification and formal measurement; the appendix documents the audited recovery of a freeze-hash metadata defect. Both Always-Spec and Guard passed 48/48 direct-token comparisons in qualification.

The prospective M4 study runs 3 repetitions of target-only, Always-Spec, and Guard on each held-out prompt, randomizing condition order within prompt and repetition: 540 observations. Both speculative conditions preserve 180/180 complete direct-ID sequences, with zero mismatches. For each prompt and condition, analysis takes the median latency across repetitions, then the geometric mean of prompt latency ratios. This ratio-of-medians estimator differs from the historical M4 text-path hardware estimator. Confidence intervals use 10,000 prompt bootstrap resamples with seed 20260924; repetitions are not independent sampling units.

Guard is profitable against target-only on this held-out workload (Table 3). Fixed speculation has the higher aggregate speedup point estimate. A reviewer-requested post-study paired secondary analysis of the frozen prompt medians gives Guard/Always-Spec latency 1.0271 [1.0130, 1.0436]. The interval lies above parity: Guard had higher latency on this comparison. By prompt median, Guard is faster

on 24 prompts, Always-Spec on 36, with 0 ties. Guard falls back in 30.6% of its 180 request executions; this measures control activity, not the fraction of beneficial decisions. Guard's geometric-mean latency relative to the offline oracle is 1.0278×. That oracle chooses the lower target-only or Always-Spec median after measurement and was unavailable to the runtime. It is a descriptive hindsight comparator, not an achievable-policy guarantee.

Guard preserves direct-ID output and is profitable against target-only, but the post-study paired comparison gives no incremental benefit over Always-Spec. Neither fallback activity nor the oracle ratio establishes risk reduction or threshold optimality. The experiment supplies no evidence of policy transfer to another stack. Small individual Spec-Bench strata do not support stable subgroup estimates. No runtime energy claim is made. The separate appendix documents the archived recomputation and audited freeze-hash metadata exception.

8 Related Work

Performance and heterogeneous execution. Speculative decoding amortizes target computation under its algorithmic assumptions (Leviathan et al., 2023). Prior work already documents verification and workload limits: Liu et al. (2026b) study verification dominance and batch effects in vLLM; Kim et al. (2026) account for conditional execution cost and runtime control; Lyu et al. (2026) derive edge–cloud profitability conditions. On consumer Apple silicon, Chordiya (2026) report gains and slowdowns across five backend configurations, diagnose ineffective batch-parallel verification in one quantized Metal path, test output agreement, and observe target-only batch-shape argmax changes. Slowdown and correctness gating are therefore established background. Heterogeneous placement is also established: Dovetail, DuoDecoding, and Mirror-SD study GPU/CPU or GPU/NPU paths (Zhang et al., 2025; Lv et al., 2025; Bhendawade et al., 2025); Ruiz y Mesa et al. (2026) combine an analytical cost model with compiler-oriented CPU/GPU partitioning, and Lever jointly optimizes draft, verification, and CPU/NPU execution for a smartphone target (Wang et al., 2026). State reuse has prior art in compressed draft KV memory (Yuan et al., 2026). Our 358H evidence instead accounts for one measured, combined replay-plus-proposal interface; it does not claim a general placement or persistence result.

Numerical paths and output contracts. Zhang et al. (2026) distinguish ragged-batch synchronization errors from residual numerical mismatches. Yuan et al. (2025) study output variation across system configurations, and LLM-42 addresses deterministic inference through verified speculation (Gond et al., 2026). Numerical path dependence is not new here. Our distinct test prospectively qualifies specific RTX/B570 paths against captured generated IDs, separately from historical text-path timing. For B570, H1 was selected after 36/36 exposed discovery matches, then complete predictions were frozen before 72 reserved live executions; four prompts could have contradicted a reference-echo explanation and instead matched the non-reference prediction in all 12 runs. H2 was unrun in that original B570 tree. Subsequent same-state B570 forks test sampled immediate and continuation effects separately; RTX's failed H1 and independently frozen P/R replay study supply a different program-level prospective test on shared numerical routines. This tests path admissibility and predictive sufficiency, not a new numerical phenomenon, unique kernel cause, or competing reserved predictor.

Adaptive speculation and the study boundary. TurboSpec adjusts speculation for serving goodput (Liu et al., 2025); BanditSpec selects configurations online (Hou et al., 2025); ACL HeteroSpec adapts verification from contextual entropy (Liu et al., 2026a). Adaptive control is established prior work. Our M4 Guard is a qualified runtime control test, not a new policy advantage: it beats target-only, but fixed speculation is faster in the post-study paired secondary comparison. The paper's central evidence is prospective direct-ID qualification, bounded frozen trajectory prediction and intervention, and same-path realized-cost evaluation.

9 Limitations

These consumer-system cases use a narrow Qwen3 family; hardware, OS, backend, versions and some quantization co-vary. They do not establish architectural rankings, population failure rates, production-load behavior or a cause for M4's separate-session contrast. The strict contract is conditional on each frozen reference. Historical M4/RTX/B570 text-path counts cannot certify direct-ID timing; Figure 3's original qualification campaigns remain unmatched to latency. The new RTX cost study qualifies only its own instrumented frontend and boundary, without qualified batched timing or uninstrumented deployment performance.

Original B570 H1 covers one stack and 24 reserved prompts, four reference-divergent. Its new intervention covers one frozen sampled checkpoint per run: immediate decisions are unchanged despite score and later-trajectory effects. Continuation remains at alternate UB128, so it does not isolate a single-block restoration. RTX H1 fails discovery; subsequent H2 is an independently orchestrated program sharing numerical routines, not B570's draft-free predictor or adversarial OS isolation. Its 72/72 prediction result coexists with 60/72 reference failure. Broader models, contexts, kernels, samplers and backend transfer remain untested. Functional HumanEval execution and pass@k were not evaluated.

Prompt bootstraps exclude full session/thermal uncertainty; repeated runs and decision events are correlated. No-known-exposure applies to frozen audited records, not inaccessible host files. Git content hashes and server/host chronology have different evidential roles; sanitized ordinals and re-hashed anonymous content do not authenticate original remote times or omitted external binaries. Source ZIPs rebuild documents; reviewer evidence packages recompute retained observations.

358H replay and proposal remain inseparable; persistence benefit, formal runtime performance and end-to-end energy remain unmeasured. Its source is `STRONGLY_SUPPORTED`, with runtime diagnostics available only as compact records without raw-trace replay. Energy domains cannot support a common ranking; M4's corrected synthetic ceiling remains unexecuted.

M4 Guard's paired comparison is post-study secondary analysis, not the registered primary endpoint. Guard had higher latency than Always-Spec; uncertainty excludes session and recalibration variation. The separate Guard/oracle ratio is hindsight regret, not that paired comparison or a policy guarantee. Historical outcomes informed design, and the audited freeze-hash metadata defect retains its original markers and STOPPED state. Control transfer, stable small strata and runtime energy benefits remain unestablished.

10 Conclusion

Output qualification, frozen prediction and realized cost answer different questions. Original direct-ID gates qualify RTX serialization and leave B570 blocked. On B570, frozen target-only UB1 predicts 72/72 capped runs across 24 repeat-stable prompts, including four reference-divergent prompts (12 runs) versus reference/live 60/72. A new same-state intervention changes scores and some later trajectories but no immediate sampled decisions. Separately frozen RTX H2 predicts 72/72 on another 24-prompt set while its UB1 reference matches 60/72; shared numerical routines and draft-dependent prediction bound that replication. New freshly qualified RTX serialization is slower ($0.646\times$) within its instrumented boundary. On 358H, combined replay/reconstruction plus proposal consumes isolated draft advantage without measured persistence payoff. M4 Guard preserves IDs and beats target-only ($1.3114\times$), but fixed speculation ($1.3469\times$) wins the post-study paired comparison; no incremental control benefit is shown. No unique numerical cause or portable performance law follows.

REFERENCES

Bhendawade, N., Nishu, K., Kundu, A., Bartels, C., Cho, M., and Belousova, I. Mirror speculative decoding: Breaking the serial barrier in LLM inference. *arXiv preprint arXiv:2510.13161v2*, 2025. URL https://arxiv.org/abs/2510.13161v2.

Chen, M., Tworek, J., Jun, H., Yuan, Q., Ponde de Oliveira Pinto, H., Kaplan, J., Edwards, H., Burda, Y., Joseph, N., Brockman, G., Ray, A., Puri, R., Krueger, G., Petrov, M., Khlaaf, H., Sastry, G., Mishkin, P., Chan, B., Gray, S., Ryder, N., Pavlov, M., Power, A., Kaiser, L., Bavarian, M., Winter, C., Tillet, P., Petroski Such, F., Cummings, D., Plappert, M., Chantzis, F., Barnes, E., Herbert-Voss, A., Hebgen Guss, W., Nichol, A., Paino, A., Tezak, N., Tang, J., Babuschkin, I., Balaji, S., Jain, S., Saunders, W., Hesse, C., Carr, A. N., Leike, J., Achiam, J., Misra, V., Morikawa, E., Radford, A., Knight, M., Brundage, M., Murati, M., Mayer, K., Welinder, P., McGrew, B., Amodei, D., McCandlish, S., Sutskever, I., and Zaremba, W. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374v2*, 2021. URL https://arxiv.org/abs/2107.03374v2.

Chordiya, P. Lossless but not free: An empirical anatomy of speculative decoding on consumer hardware. *arXiv preprint arXiv:2607.17283*, 2026. doi: 10.48550/arXiv.2607.17283. URL https://arxiv.org/abs/2607.17283.

Gond, R., Kamath, A. K., Ramjee, R., and Panwar, A. LLM-42: Enabling determinism in LLM inference with verified speculation. *arXiv preprint arXiv:2601.17768v2*, 2026. URL https://arxiv.org/abs/2601.17768v2.

Hou, Y., Zhang, F., Du, C., Zhang, X., Pan, J., Pang, T., Du, C., Tan, V., and Yang, Z. BanditSpec: Adaptive speculative decoding via bandit algorithms. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267, pp. 24045–24079. PMLR, 2025. URL https://proceedings.mlr.press/v267/hou25h.html.

Kim, S., Kim, J., Cho, Y., and Seo, J. Runtime-controlled evaluation of speculative decoding methods. *Electronics*, 15(14):3179, 2026. doi: 10.3390/electronics15143179. URL https://www.mdpi.com/2079-9292/15/14/3179.

Leviathan, Y., Kalman, M., and Matias, Y. Fast inference from transformers via speculative decoding. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202, pp. 19274–19286. PMLR, 2023. URL https://proceedings.mlr.press/v202/leviathan23a.html.

Liu, S., Ye, Y., Zhu, Q., Cao, Z., and He, Y. HeteroSpec: Leveraging contextual heterogeneity for efficient speculative decoding. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 12930–12947. Association for Computational Linguistics, July 2026a. doi: 10.18653/v1/2026.acl-long.589. URL https://aclanthology.org/2026.acl-long.589/.

Liu, X., Park, J., Hu, L., Kwon, W., Li, Z., Zhang, C., Du, K., Mo, X., You, K., Cheung, A., Deng, Z., Stoica, I., and Zhang, H. TurboSpec: Closed-loop speculation control system for optimizing LLM serving goodput. *arXiv preprint arXiv:2406.14066v3*, 2025. URL https://arxiv.org/abs/2406.14066v3.

Liu, X., Yu, J., Park, J., Stoica, I., and Cheung, A. Speculative decoding: Performance or illusion? In *Proceedings of Machine Learning and Systems*, volume 8, pp. 1719–1741, 2026b. URL https://proceedings.mlsys.org/paper_files/paper/2026/hash/554e056fe2b6d9fd27ffcd3367ae1267-Abstract-Conference.html.

Lv, K., Guo, H., Guo, Q., and Qiu, X. DuoDecoding: Hardware-aware heterogeneous speculative decoding with dynamic multi-sequence drafting. *arXiv preprint arXiv:2503.00784*, 2025. URL https://arxiv.org/abs/2503.00784.

Lyu, Y., Irukulapati, B., and Champati, J. P. Speculation at a distance: Where edge-cloud speculative decoding actually pays off. *arXiv preprint arXiv:2606.25091v2*, 2026. URL https://arxiv.org/abs/2606.25091v2.

Ruiz y Mesa, A., Korol, G., Riesterer, M., Cardoso de Lima, J. P., and Castrillon, J. Compiler-assisted speculative sampling for accelerated LLM inference on heterogeneous edge devices. In *8th Workshop on Accelerated Machine Learning (AccML), co-located with HiPEAC*, January 2026. URL https://accml.dcs.gla.ac.uk/papers/2026/8th_AccML_paper_8.pdf.

Wang, T., Li, F., Sun, Y., Gao, W., and Ren, J. Lever: Speculative LLM inference on smartphones. *arXiv preprint arXiv:2605.16786*, 2026. doi: 10.48550/arXiv.2605.16786. URL https://arxiv.org/abs/2605.16786.

Xia, H., Yang, Z., Dong, Q., Wang, P., Li, Y., Ge, T., Liu, T., Li, W., and Sui, Z. Unlocking efficiency in large language model inference: A comprehensive survey of speculative decoding. In *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 7655–7671. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.findings-acl.456. URL https:

//aclanthology.org/2024.findings-acl.456/.

Yuan, J., Li, H., Ding, X., Xie, W., Li, Y.-J., Zhao, W., Wan, K., Shi, J., Hu, X., and Liu, Z. Understanding and mitigating numerical sources of nondeterminism in LLM inference. *arXiv preprint arXiv:2506.09501v2*, 2025. URL https://arxiv.org/abs/2506.09501v2.

Yuan, T., Liao, C., and Wen, Z. Strong drafts need compact memories: Long-context speculative decoding with compressed KV cache. *arXiv preprint arXiv:2608.30252*, 2026. URL https://arxiv.org/abs/2608.30252. EMNLP 2026 Findings.

Zhang, L., Zhang, Z., Xu, B., Li, R., Tian, Z., Mei, S., and Li, D. Dovetail: A CPU/GPU heterogeneous speculative decoding for LLM inference. *arXiv preprint arXiv:2412.18934v2*, 2025. URL https://arxiv.org/abs/2412.18934v2.

Zhang, R. H., Dey, S., Mishra, A., Wu, H., Li, B., and Zhang, R. Correctness forensics for batch speculative decoding: Diagnosing the ragged tensor problem. *arXiv preprint arXiv:2510.22876v4*, 2026. URL https://arxiv.org/abs/2510.22876v4. Accepted to Findings of EMNLP 2026.